



Inequality Proof by Induction

Practice Question Set

Proving Inequality Statements using the Principle of
Mathematical Induction

8 Questions • Fully Worked Answers

Instructions: Attempt each proof yourself first. The full worked answer is given directly beneath each question, boxed in yellow, so you can check your working immediately.

Question 1

Prove that for all $n \in \mathbb{Z}^+$:

$$2^{-n} < \frac{1}{n}.$$

Answer

Let P_n be the proposition: $2^{-n} < \frac{1}{n}$ for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: LHS = $2^{-1} = \frac{1}{2}$ and RHS = $\frac{1}{1} = 1$. Since $\frac{1}{2} < 1$,
 $\therefore P_1$ is true.

(2) If P_k is true, then $2^{-k} < \frac{1}{k}$ for some $k \geq 1$.

Now

$$\begin{aligned} 2^{-(k+1)} &= 2^{-k} \times \frac{1}{2} \\ &< \frac{1}{k} \times \frac{1}{2} && \{\text{using } P_k\} \\ &< \frac{1}{k+1} && \{k \geq 1, \text{ so } 2k \geq k+1\}. \end{aligned}$$

$\therefore P_{k+1}$ is also true. Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 2

Prove that for all integers $n \geq 0$:

$$3^n \geq 1 + 2n.$$

Answer

Let P_n be the proposition: $3^n \geq 1 + 2n$ for integers $n \geq 0$.

Proof: (By the principle of mathematical induction)

(1) If $n = 0$: LHS = $3^0 = 1$ and RHS = $1 + 2(0) = 1$. Since $1 \geq 1$, $\therefore P_0$ is true.

(2) If P_k is true, then $3^k \geq 1 + 2k$.

Now

$$\begin{aligned} 3^{k+1} &= 3^k \times 3 \geq (1 + 2k) \times 3 && \{\text{using } P_k\} \\ &\geq 3 + 6k \\ &\geq 3 + 2k && \{k \geq 0\} \\ &\geq 1 + 2(k + 1). \end{aligned}$$

$\therefore P_{k+1}$ is also true. Since P_0 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all integers $n \geq 0$. {principle of mathematical induction}

Question 3

Prove that for all $n \in \mathbb{Z}, n \geq 4$:

$$n! \geq 2^n.$$

Answer

Let P_n be the proposition: $n! \geq 2^n$ for all $n \in \mathbb{Z}, n \geq 4$.

Proof: (By the principle of mathematical induction)

(1) If $n = 4$: LHS = $4! = 24$ and RHS = $2^4 = 16$. Since $24 \geq 16$,
 $\therefore P_4$ is true.

(2) If P_k is true, then $k! \geq 2^k$.

Now

$$\begin{aligned} (k+1)! &= (k+1) \times k! \geq (k+1) \times 2^k && \{\text{using } P_k\} \\ &\geq 2 \times 2^k && \{k \geq 4\} \\ &\geq 2^{k+1}. \end{aligned}$$

$\therefore P_{k+1}$ is also true. Since P_4 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}, n \geq 4$.
 {principle of mathematical induction}

Question 4

Prove that for all $n \in \mathbb{Z}^+$:

$$8^n \geq n^3.$$

Answer

Let P_n be the proposition: $8^n \geq n^3$ for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: LHS = $8^1 = 8$ and RHS = $1^3 = 1$. Since $8 \geq 1$,
 $\therefore P_1$ is true.

(2) If P_k is true, then $8^k \geq k^3$.

Now

$$\begin{aligned} 8^{k+1} &= 8 \times 8^k \geq 8 \times k^3 && \{\text{using } P_k\} \\ &\geq (2k)^3 \\ &\geq (k+1)^3 && \{k \geq 1, \text{ so } 2k \geq k+1\}. \end{aligned}$$

$\therefore P_{k+1}$ is also true. Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 5

Prove that for all $n \in \mathbb{Z}^+$:

$$\sum_{i=1}^n \frac{1}{i} \leq \frac{n}{2} + 1.$$

Answer

Let P_n be the proposition: $\sum_{i=1}^n \frac{1}{i} \leq \frac{n}{2} + 1$ for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: LHS = $\frac{1}{1} = 1$ and RHS = $\frac{1}{2} + 1 = \frac{3}{2}$. Since $1 \leq \frac{3}{2}$,
 $\therefore P_1$ is true.

(2) If P_k is true, then $\sum_{i=1}^k \frac{1}{i} \leq \frac{k}{2} + 1$.

Now

$$\begin{aligned} \sum_{i=1}^{k+1} \frac{1}{i} &= \left(\sum_{i=1}^k \frac{1}{i} \right) + \frac{1}{k+1} \\ &\leq \frac{k}{2} + \frac{1}{k+1} + 1 \quad \{\text{using } P_k\}. \end{aligned}$$

and

$$\begin{aligned} \frac{k}{2} + \frac{1}{k+1} - \frac{k+1}{2} &= \frac{1}{k+1} - \frac{1}{2} \\ &= \frac{2 - (k+1)}{2(k+1)} \\ &= \frac{1-k}{2(k+1)} \leq 0 \text{ for all } k \in \mathbb{Z}^+. \end{aligned}$$

$$\therefore \frac{k}{2} + \frac{1}{k+1} \leq \frac{k+1}{2}, \text{ so } \sum_{i=1}^{k+1} \frac{1}{i} \leq \frac{k+1}{2} + 1.$$

$\therefore P_{k+1}$ is also true. Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 6

Prove that for all $n \in \mathbb{Z}^+, n \geq 8$:

$$2^n > 3n^2 + 3n + 1.$$

Answer

Let P_n be the proposition: $2^n > 3n^2 + 3n + 1$ for all $n \in \mathbb{Z}^+, n \geq 8$.

Proof: (By the principle of mathematical induction)

(1) If $n = 8$: LHS = $2^8 = 256$ and RHS = $3(8)^2 + 3(8) + 1 = 217$.

Since $256 > 217$, $\therefore P_8$ is true.

(2) If P_k is true, then $2^k > 3k^2 + 3k + 1$.

Now

$$\begin{aligned} 2^{k+1} &= 2 \times 2^k > 2(3k^2 + 3k + 1) && \{\text{using } P_k\} \\ &> 6k^2 + 6k + 2. \end{aligned}$$

and

$$\begin{aligned} &6k^2 + 6k + 2 - [3(k+1)^2 + 3(k+1) + 1] \\ &= 6k^2 + 6k + 2 - 3k^2 - 6k - 3 - 3k - 3 - 1 \\ &= 3k^2 - 3k - 5 \\ &> 0 \text{ for all } k \in \mathbb{Z}^+, k \geq 8. \end{aligned}$$

$\therefore 6k^2 + 6k + 2 > 3(k+1)^2 + 3(k+1) + 1$, so $2^{k+1} > 3(k+1)^2 + 3(k+1) + 1$.

$\therefore P_{k+1}$ is also true. Since P_8 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+, n \geq 8$. {principle of mathematical induction}

Question 7

[Maximum mark: 7] Prove by mathematical induction that

$$\binom{2n}{n} \leq 2^{2n-1}, \quad \text{where } n \in \mathbb{Z}^+.$$

Answer

Let P_n be the proposition: $\binom{2n}{n} \leq 2^{2n-1}$ for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: LHS = $\binom{2}{1} = 2$ and RHS = $2^1 = 2$. Since $2 \leq 2$,
 $\therefore P_1$ is true.

(2) If P_k is true, then $\binom{2k}{k} \leq 2^{2k-1}$.

Key fact: $\binom{2k+2}{k+1} = \binom{2k}{k} \times \frac{2(2k+1)}{k+1}$.

Now

$$\begin{aligned} \binom{2k+2}{k+1} &= \binom{2k}{k} \times \frac{2(2k+1)}{k+1} \\ &\leq \binom{2k}{k} \times 4 \quad \left\{ \frac{2(2k+1)}{k+1} \leq 4 \right\} \\ &\leq 2^{2k-1} \times 4 \quad \{\text{using } P_k\} \\ &= 2^{2k+1} = 2^{2(k+1)-1}. \end{aligned}$$

since $\frac{2(2k+1)}{k+1} \leq 4 \iff 2k+1 \leq 2k+2$, which always holds.

$\therefore P_{k+1}$ is also true. Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 8

[Extra practice] [Maximum mark: 7] Prove by mathematical induction that

$$\binom{2n}{n} \geq 2^n, \quad \text{where } n \in \mathbb{Z}^+, n \geq 2.$$

Answer

Let P_n be the proposition: $\binom{2n}{n} \geq 2^n$ for all $n \in \mathbb{Z}^+, n \geq 2$.

Proof: (By the principle of mathematical induction)

(1) If $n = 2$: LHS = $\binom{4}{2} = 6$ and RHS = $2^2 = 4$. Since $6 \geq 4$,
 $\therefore P_2$ is true.

(2) If P_k is true, then $\binom{2k}{k} \geq 2^k$ for $k \geq 2$.

Key fact: $\binom{2k+2}{k+1} = \binom{2k}{k} \times \frac{2(2k+1)}{k+1}$.

Now

$$\begin{aligned} \binom{2k+2}{k+1} &= \binom{2k}{k} \times \frac{2(2k+1)}{k+1} \\ &\geq \binom{2k}{k} \times 2 \quad \left\{ \frac{2(2k+1)}{k+1} \geq 2 \right\} \\ &\geq 2^k \times 2 \quad \{\text{using } P_k\} \\ &= 2^{k+1}. \end{aligned}$$

since $\frac{2(2k+1)}{k+1} \geq 2 \iff 2k+1 \geq k+1 \iff k \geq 0$, which
always holds.

$\therefore P_{k+1}$ is also true. Since P_2 is true, and P_{k+1} is true whenever P_k
is true, P_n is true for all $n \in \mathbb{Z}^+, n \geq 2$. {principle of
mathematical induction}

Question 7 Alternative method

[Maximum mark: 7] Prove by mathematical induction that

$$\binom{2n}{n} \leq 2^{2n-1}, \quad \text{where } n \in \mathbb{Z}^+.$$

Answer

Let P_n be the proposition: $\binom{2n}{n} \leq 2^{2n-1}$ for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: LHS = $\binom{2}{1} = 2$ and RHS = $2^1 = 2$. Since $2 \leq 2$, $\therefore P_1$ is true.

(2) Assume P_k is true, that is $\binom{2k}{k} \leq 2^{2k-1}$, i.e. $\frac{(2k)!}{k! k!} \leq 2^{2k-1}$.

Now P_{k+1} is $\binom{2(k+1)}{k+1} \leq 2^{2(k+1)-1}$.

$$\begin{aligned} \binom{2k+2}{k+1} &= \frac{(2k+2)!}{(k+1)!(k+1)!} \\ &= \frac{(2k+2)(2k+1)(2k)!}{(k+1)k!(k+1)k!} \\ &= \frac{2(k+1)(2k+1)}{(k+1)(k+1)} \times \frac{(2k)!}{k!k!} = \frac{2(2k+1)}{k+1} \times \frac{(2k)!}{k!k!} \\ &\leq \frac{2(2k+1)}{k+1} \times 2^{2k-1} \quad \{\text{using } P_k\} \\ &= \frac{2k+1}{k+1} \times 2^{2k} \\ &\leq 2 \times 2^{2k} \quad \left\{ \frac{2k+1}{k+1} \leq 2 \text{ for } k \geq 1 \right\} \\ &= 2^{2k+1} = 2^{2(k+1)-1}. \end{aligned}$$

$\therefore P_{k+1}$ is also true. Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 8

Alternative method

[Extra practice] [Maximum mark: 7] Prove by mathematical induction that

$$\binom{2n}{n} \geq 2^n, \quad \text{where } n \in \mathbb{Z}^+, n \geq 2.$$

Answer

Let P_n be the proposition: $\binom{2n}{n} \geq 2^n$ for all $n \in \mathbb{Z}^+, n \geq 2$.

Proof: (By the principle of mathematical induction)

(1) If $n = 2$: LHS = $\binom{4}{2} = 6$ and RHS = $2^2 = 4$. Since $6 \geq 4$, $\therefore P_2$ is true.

(2) Assume P_k is true, that is $\binom{2k}{k} \geq 2^k$, i.e. $\frac{(2k)!}{k!k!} \geq 2^k$, for $k \geq 2$.

Now P_{k+1} is $\binom{2(k+1)}{k+1} \geq 2^{k+1}$.

$$\begin{aligned} \binom{2k+2}{k+1} &= \frac{(2k+2)!}{(k+1)!(k+1)!} \\ &= \frac{(2k+2)(2k+1)(2k)!}{(k+1)k!(k+1)k!} \\ &= \frac{2(k+1)(2k+1)}{(k+1)(k+1)} \times \frac{(2k)!}{k!k!} = \frac{2(2k+1)}{k+1} \times \frac{(2k)!}{k!k!} \\ &\geq \frac{2(2k+1)}{k+1} \times 2^k \quad \{\text{using } P_k\} \\ &= \frac{2k+1}{k+1} \times 2^{k+1} \\ &\geq 1 \times 2^{k+1} \quad \left\{ \frac{2k+1}{k+1} \geq 1 \text{ for } k \geq 0 \right\} \\ &= 2^{k+1}. \end{aligned}$$

$\therefore P_{k+1}$ is also true. Since P_2 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+, n \geq 2$. {principle of mathematical induction}